4.5 System Deployment & Integration

Platform: **EduHub** (Online Learning Platform) | Track: Full Stack .NET Web Development

1. Technology Stack

Architectural Layer	Adopted Technologies & Frameworks
Presentation Layer	ASP.NET Core MVC (Razor Views), Bootstrap 5.3, Video.js (custom video player), Chart.js (interactive metrics), DataTables (administrative grid management), jQuery & jQuery Validation.
Business Logic Layer	C# (.NET 8 LTS), FluentValidation, Strategy Pattern (promotional/discount policies), Observer Pattern (progress evaluation & certificate issuance).
Data Access Layer	Entity Framework Core (Code-First with Fluent API), Microsoft SQL Server / Azure SQL Database, Generic Repository & Unit of Work patterns.
Security & Identity	ASP.NET Core Identity with Role-Based Access Control (Student, Instructor, Admin), Claim-based authorization filters, anti-forgery protection.
Cloud & Integrations	Azure App Service (Linux runtime), Azure Blob Storage (SAS token streaming), Azure CDN, Stripe Checkout API (payments), iTextSharp (PDF certification).

2. Deployment Diagram (Cloud Infrastructure Architecture)

EduHub leverages a secure, highly available cloud hosting topology on Microsoft Azure:



3. Component Diagram (Internal Software Decomposition)

